Criteo Uplift Benchmark Report

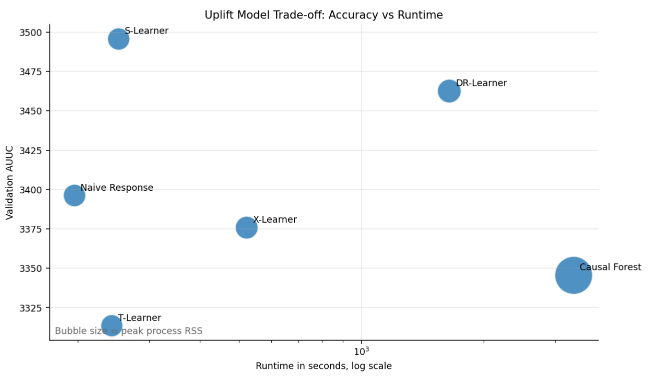
Sample report for campaign incrementality modeling. The benchmark asks which users convert because of treatment, not only which users are likely to convert.

ROWS	BEST TRADEOFF	TEST AUUC	TOP DECILE
7M	S-Learner	3405.48	6.23x

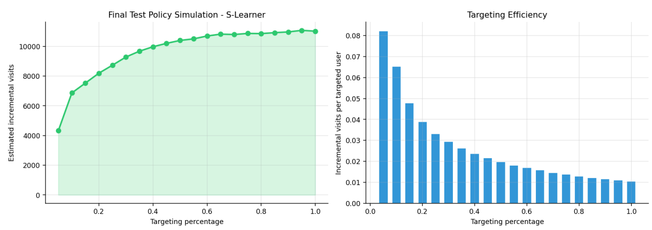
Model Comparison

Model	Validation AUUC	Runtime	Peak RSS
S-Learner	3495.68	252.3s	5041 MB
DR-Learner	3462.59	1646.7s	5727 MB
Naive response ranker	3396.22	196.4s	5113 MB
X-Learner	3375.79	521.7s	5288 MB
Causal Forest	3345.46	3335.2s	14458 MB

Accuracy vs Runtime



Policy Simulation



Business Read

- Use uplift ranking for campaign targeting, not raw conversion probability.
- S-Learner is the practical winner in this run: strong AUUC with much lower runtime than DR-Learner and Causal Forest.
- The top decile shows a 6.23x relative uplift signal, which is the first segment to test in a controlled campaign rollout.
- The paired bootstrap interval overlaps zero, so this should be treated as a production trade-off decision rather than a universal model claim.

Delivered Artifacts

Artifact	Purpose
benchmark_results.csv	AUUC, Qini, runtime, and memory by model
policy_segments.csv	Treatment targeting groups and expected incremental response
diagnostic_charts/	Qini curve, uplift by decile, SHAP surrogate, policy simulation
README assets	Repo-ready visuals for portfolio and stakeholder review